Chapter 2 Exploring Data with Tables and Graphs
Graphs that Enlighten and Graphs that Deceive (2.3)

Objectives

In this lesson, we will learn how to

- recognize several common graphs that help us understand data;
- choose appropriate graphs for quantitative, categorical, and time-series data;
- identify misleading features in graphs;
- explain basic principles for making graphs clear and honest.

1. Why More Graphs?

A graph can help us see patterns that are difficult to notice in a table of numbers.

Key Idea

Good graphs should enlighten us. They should reveal the true nature of the data instead of distracting or misleading the reader.

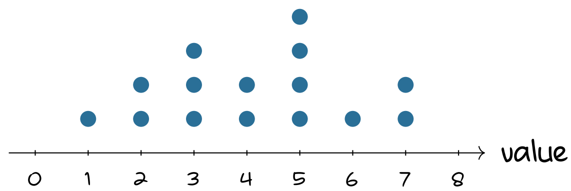
In this section, we consider graphs that are useful and graphs that can be deceptive.

2. Dotplots

Dotplots display quantitative data by plotting each data value as a dot above a horizontal scale. If values repeat, the dots are stacked.

Dotplot

A dotplot is a graph of quantitative data in which each data value is plotted as a point above a horizontal scale of values.



Why Dotplots Are Helpful

- They show the shape of the distribution.
- They often allow us to recreate the original list of data values.

Example 1: A small data set has fewer than 20 values. Would a dotplot be a reasonable graph to use? Explain.

Solution

Yes. A dotplot is very useful for a small quantitative data set because it shows the shape of the distribution and usually preserves the original data values.

3. Stemplots

A stemplot, also called a stem-and-leaf plot, separates each value into two parts: a stem and a leaf.

Stemplot

A stemplot represents quantitative data by separating each value into a stem and a leaf.

For example, the number 47 can be written as

$$4 \mid 7,$$

where 4 is the stem and 7 is the leaf.

Why Stemplots Are Helpful

- They show the shape of the distribution.
- They retain the original data values.
- They sort the data values in order.

Example 2: Create a stemplot for the data values

$$21, 24, 26, 31, 33, 34, 42.$$

Solution

One possible stemplot is

$$\begin{array}{r|l} 2 & 1\ 4\ 6 \\ 3 & 1\ 3\ 4 \\ 4 & \ \ 2 \end{array}$$

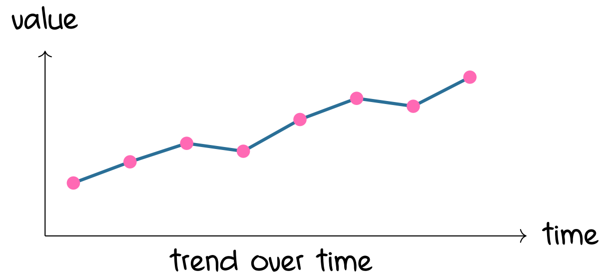
The stems are the tens digits, and the leaves are the ones digits.

4. Time-Series Graphs

A time-series graph is used when quantitative data are collected at different points in time, such as days, months, or years.

Time-Series Graph

A time-series graph displays data collected over time and is especially useful for revealing trends.



Example 3: Suppose we record the number of students enrolled in an online course each semester for several years. What type of graph would be appropriate?

Solution

A time-series graph would be appropriate because the data are collected over time, and we want to see whether enrollment is increasing, decreasing, or staying about the same.

5. Bar Graphs and Pareto Charts

A bar graph is used for categorical data. The bars show the frequency or relative frequency of each category.

Bar Graph

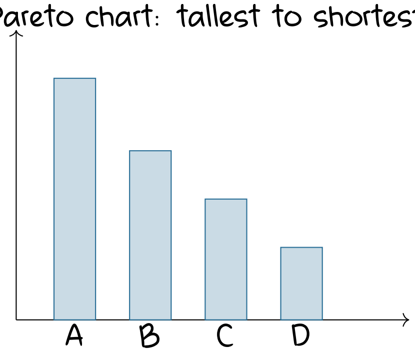
A bar graph uses bars of equal width to show frequencies of categories of categorical data.

A Pareto chart is a special type of bar graph in which the bars are arranged from tallest to shortest.

Pareto Chart

A Pareto chart is a bar graph for categorical data with bars arranged in descending order according to frequency.

Pareto chart: tallest to shortest



Why Pareto Charts Are Helpful

Pareto charts draw attention to the categories with the largest frequencies. This helps us focus on the most important categories first.

6. Pie Charts

A pie chart shows categorical data as slices of a circle.

Pie Chart

A pie chart depicts categorical data as slices of a circle, where the size of each slice is proportional to the frequency of the category.

Pie charts are common, but they can be harder to read accurately than bar graphs, especially when the slices are similar in size.

Example 4: For comparing the number of students in several majors, would a bar graph or pie chart usually be easier to read?

Solution

A bar graph is usually easier to read because the heights of bars are easier to compare than the areas or angles of slices in a pie chart.

7. Frequency Polygons

A frequency polygon is similar to a histogram, but it uses points and line segments instead of bars.

Frequency Polygon

A frequency polygon uses line segments connected to points located above class midpoint values.

A relative frequency polygon uses relative frequencies or percentages on the vertical scale.

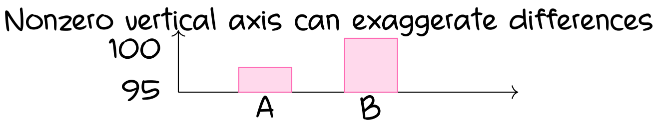
Frequency polygons are useful when comparing two or more distributions on the same graph.

8. Graphs That Deceive

Some graphs create impressions that are misleading or wrong.

Warning: Nonzero Vertical Axis

A graph can exaggerate differences if the vertical axis starts at a number greater than zero.



Always check whether the vertical axis begins at zero. If it does not, the graph may make small differences look much larger than they really are.

9. Pictographs

Pictographs use pictures or drawings to represent data. They may look interesting, but they can easily be misleading.

Why Pictographs Can Mislead

When one-dimensional data are represented with two-dimensional or three-dimensional pictures, the visual impression can be distorted.

For example:

- Doubling each side of a square makes its area 4 times as large.
- Doubling each side of a cube makes its volume 8 times as large.

So a picture may look much larger even when the actual data value is only twice as large.

Example 5: A graph uses dollar-bill pictures to compare two budgets. One budget is twice as large as the other, but the picture is twice as wide and twice as tall. Why is this misleading?

Solution

If the picture is twice as wide and twice as tall, its area is four times as large. This makes the larger budget look four times as large, even though it is only twice as large. The graph exaggerates the difference.

10. Good Graph Design

Principles for Better Graphs

- For small data sets of 20 values or fewer, a table may be better than a graph.
- A graph should make us focus on the data, not on distracting design features.
- Do not distort the data.
- Most of the ink in a graph should be used for the data, not decoration.

Example 6: A colorful 3D pie chart looks impressive, but it makes some categories look larger because they are closer to the viewer. Is this a good graph? Explain.

Solution

No. A graph should reveal the true nature of the data. A 3D effect can distort the sizes of categories and make the graph harder to interpret accurately.

Summary

Key Ideas

- Dotplots and stemplots are useful for small quantitative data sets.
- Time-series graphs show trends over time.
- Bar graphs and Pareto charts are useful for categorical data.
- Pie charts are common but can be harder to compare accurately.

- Frequency polygons are useful for comparing distributions.
- Graphs can be deceptive if they use nonzero axes, pictographs, or distracting designs.
- Good graphs should reveal the true nature of the data.